

TEAM NEUTRONIC

Real-Time Data Logger for Global Transportation

Project Report · Version 1.0

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13 July 2026

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1. Introduction

1.1 Project Overview

Problem Statement

The global cold-chain logistics industry, particularly the pharmaceutical and biologics sector, faces a critical challenge: temperature excursions during transport and storage cause an estimated \$35 billion in annual product spoilage, with a single cold-chain failure potentially costing an organisation up to \$95 million in medical inventory. Regulatory frameworks such as Good Distribution Practice (GDP) mandate continuous, accurate temperature monitoring with traceable calibration, yet 92% of supply-chain professionals admit they cannot fully trust their logistics data.

Existing cold-chain monitoring solutions fall into two unsatisfactory categories:

- **Passive USB-only data loggers** (e.g., Testo 175-T2, Elitech RC-5+): record temperature locally but provide no real-time visibility. Excursions are only discovered after delivery, when intervention is impossible — essentially “post-mortem” tools.
- **Proprietary real-time trackers** (e.g., Sensitech TempTale GEO, Onset CX1003): offer cellular connectivity but are typically single-use (creating electronic waste and recurring purchase costs), lock users into expensive proprietary cloud platforms, and often have limited accuracy ($\pm 0.3^{\circ}\text{C}$ to $\pm 0.5^{\circ}\text{C}$).

Our Solution

A multi-sensor, cellular-connected logger combining high accuracy, real-time alerts, USB-C data export, and an external probe bridges the gap between cheap passive tags and expensive real-time trackers — cutting spoilage and compliance risk.

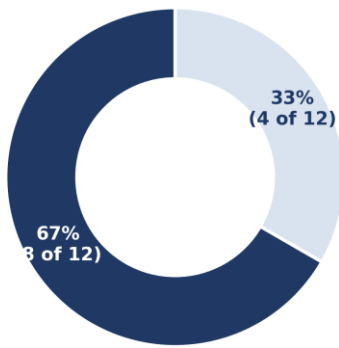
1.2 Motivation

1. **Regulatory Pressure:** Good Distribution Practice (GDP) and 21 CFR Part 11 mandate continuous, accurate temperature monitoring with traceable calibration. The latest GDP revisions require real-time monitoring capability for high-value pharmaceuticals.
2. **Market Demand:** The global cold-chain monitoring market is projected to grow from \$7.2 billion (2025) to \$22.2 billion (2035) at a 12.1% CAGR. The tracking and monitoring segment alone is expected to reach \$15.04 billion by 2030.
3. **Local Industry Survey Evidence:** A structured survey of professionals across pharmaceutical import and distribution companies in Sri Lanka confirmed the need identified above.

To validate the identified operational challenges, a targeted local survey was conducted among key healthcare stakeholders, including retail and hospital pharmacists, medical officers, and hospital administration officials.

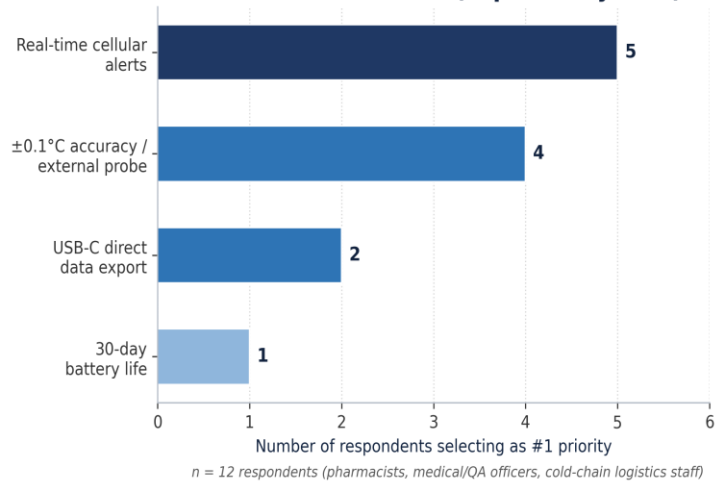
Survey Results

Have You Experienced a Cold-Chain Temperature Excursion or Product Loss?

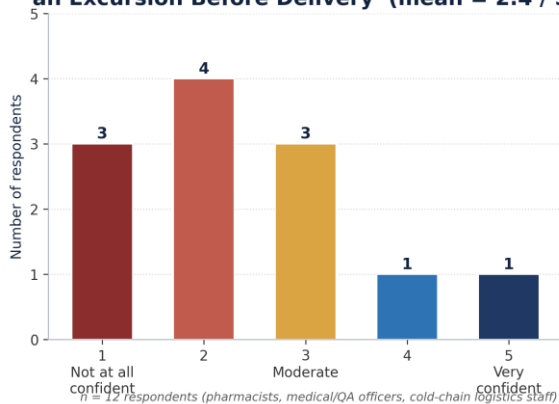


■ Experienced a spoilage / excursion incident (past 12 months)
■ No incident reported

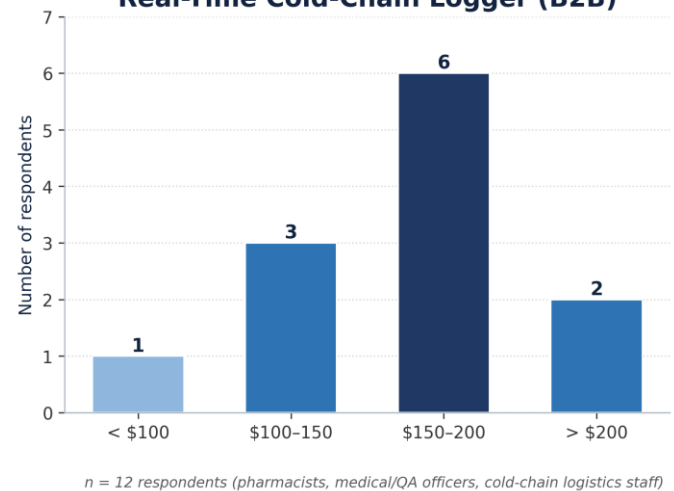
Most Valued Feature (Top Priority Pick)



Confidence That Current Monitoring Would Catch an Excursion Before Delivery (mean = 2.4 / 5)



Acceptable Unit Price for a Reusable, Real-Time Cold-Chain Logger (B2B)



1.3 Objectives

Primary Objective

Develop a reliable and scalable product to monitor temperature, humidity, acceleration, and optical light intensity of a global shipping delivery, logging information to local memory and uploading to a cloud database in real time.

Secondary Objectives

- 30 days of operating time on a single charge
- Global coverage for data upload
- User-configurable logging and upload intervals
- Local data access as a CSV file via mass-storage device
- Interrupt-based alarming system for critical changes in readings

2. Product Overview

2.1 Features

#	Feature	Specification
1	Temperature Accuracy	$\pm 0.1^{\circ}\text{C}$ (RTD sensor)
2	Humidity Accuracy	$\pm 1.0\%$ RH (SHT45)
3	External Probe	PT100 RTD via MAX31865 (-200°C to $+200^{\circ}\text{C}$)
4	Connectivity	LTE-M / NB-IoT (SIM7070G HAT), 2G fallback
5	Battery	5000 mAh 21700 Li-Ion, rechargeable, 30-day runtime
6	Data Export	USB-C mass-storage (auto-generated PDF/CSV), no drivers
7	Cloud Protocol	HTTP over TLS, open platform
8	Sensors	Light (OPT3001, door-open detection); accelerometer (LIS3DH, tilt/shock); fuel gauge (MAX17048)
9	User Interface	1.3" OLED display, 3 user buttons, buzzer
10	Enclosure	IP65, ABS, magnetic mounting, $85 \times 65 \times 25$ mm

2.2 Technical Feasibility

2.2.1 Resource Requirements

Resource	Requirement	Availability	Status
Microcontroller	Ultra-low-power, USB 2.0, ≥ 128 KB Flash	STM32L412CBT6 (LQFP-48) — LCSC, DigiKey	☒ Confirmed

Resource	Requirement	Availability	Status
Temperature/Humidity Sensor	±0.1°C accuracy, I ² C, industrial range	Sensirion SHT45-AD1B-R2	☑ Confirmed (SHT40-AD1F-R2 backup)
Cellular Module	LTE-M/NB-IoT, global bands, low-power PSM	Waveshare SIM7070G HAT	☑ Confirmed
External RTD Front-end	SPI, PT100 support, industrial temp	MAX31865AAP+ (SSOP-20)	☑ Confirmed
Memory	16 MB SPI NOR flash, >100k write cycles	Winbond W25Q128JVS1Q (SOIC-8)	⚠ Switched to W25R128JVS1Q (availability)
Battery	≥5000 mAh, rechargeable, single cell	Samsung 50E 21700 (5000 mAh)	⚠ Switched to 3600 mAh 18650 (availability)
Other Sensors	Ambient light; accelerometer; fuel gauge	TI OPT3001; ST LIS3DHTR; MAX17048	☑ Confirmed
Power	High-efficiency regulation, charging, protection, load sharing	TPS630001 buck-boost; MCP73831 charger; S-8211CAH+FS8205A protection; DMP1045U MOSFET	☑ Confirmed
PCB Fabrication	4-layer, controlled impedance (90Ω USB)	JLCPCB (JLC04161H-7628 stackup)	☑ Confirmed
Development Tools	IDE, debugger, test equipment	STM32CubeIDE, ST-LINK V2, oscilloscope	☑ Confirmed

Technical Feasibility Confirmation

The project is determined to be **technically feasible**. All critical components are commercially available from multiple suppliers (LCSC, DigiKey), ensuring supply-chain resilience.

- The STM32L412CBT6 microcontroller meets the ultra-low-power (deep-sleep modes) and USB 2.0 requirements.
- The SIM7070G HAT eliminates complex RF design by providing a pre-certified cellular solution.
- A 24-hour current measurement confirmed an average draw of 12.4 mA (initial, unconfigured), corresponding to a 12-day worst-case battery life; with configured deep-sleep modes, ~30 days of operation is achievable.
- The PT100 sensor was validated against a calibrated reference thermometer in a climate chamber, achieving the specified ±0.1°C accuracy.
- USB mass-storage enumeration and PDF generation were successful.

In summary, all resource requirements can be met, all performance targets are reachable, and the identified risks have viable mitigations — confirming the project is ready to proceed to detailed design and prototyping.

2.3 System Components

The Smart Cold-Chain Data Logger integrates the following major subsystems:

- **Sensing Board:** SHT45, OPT3001, LIS3DH, MAX17048 on a shared I²C bus; MAX31865 on SPI.

- **MCU Board:** STM32L412CBT6 (Cortex-M4, 80 MHz) with external 32.768 kHz crystal for RTC.
- **Storage:** 16 MB SPI NOR flash (W25Q128).
- **Communication:** SIM7070G HAT (UART), USB-C for power and data.
- **Power Management:** MCP73831 charger, TPS631000 buck-boost, S-8211CAH battery protection, load-sharing circuit (DMP1045U MOSFET).
- **User Interface:** 1.3" OLED (I²C), user buttons, piezo buzzer.

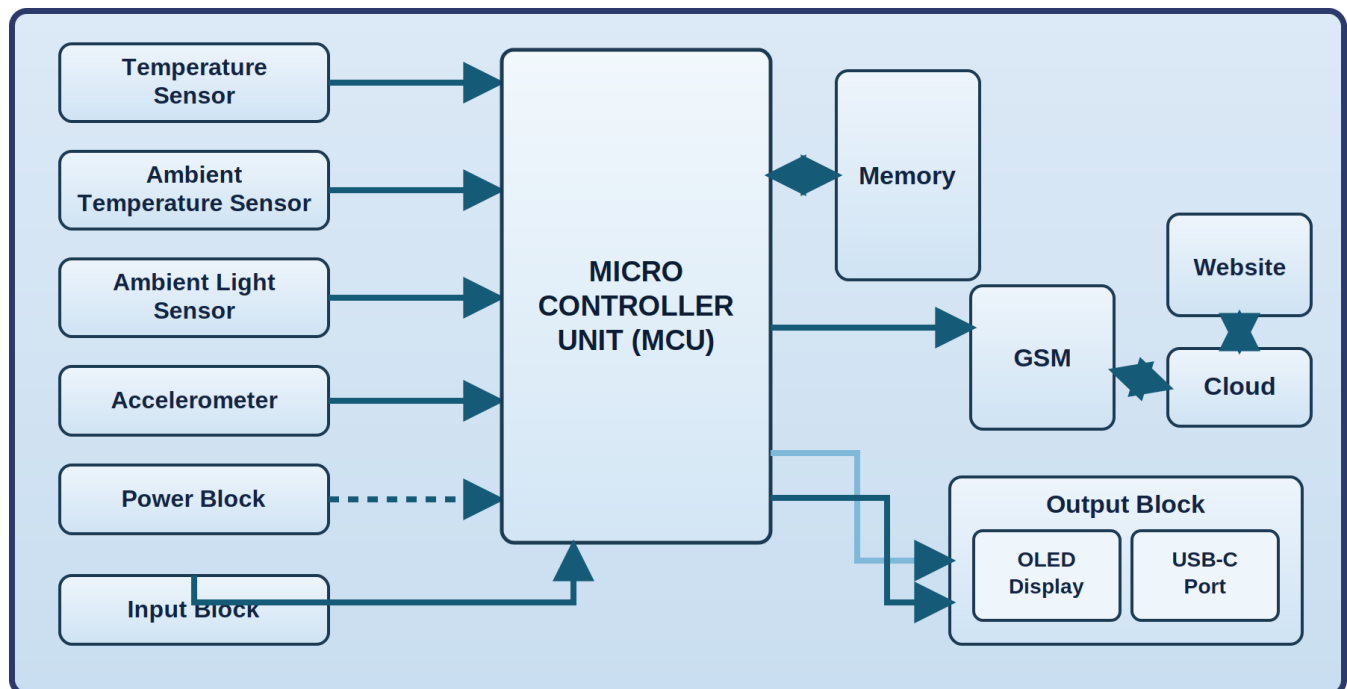
A block diagram appears in Section 3.1.

2.4 Methodology and Technology

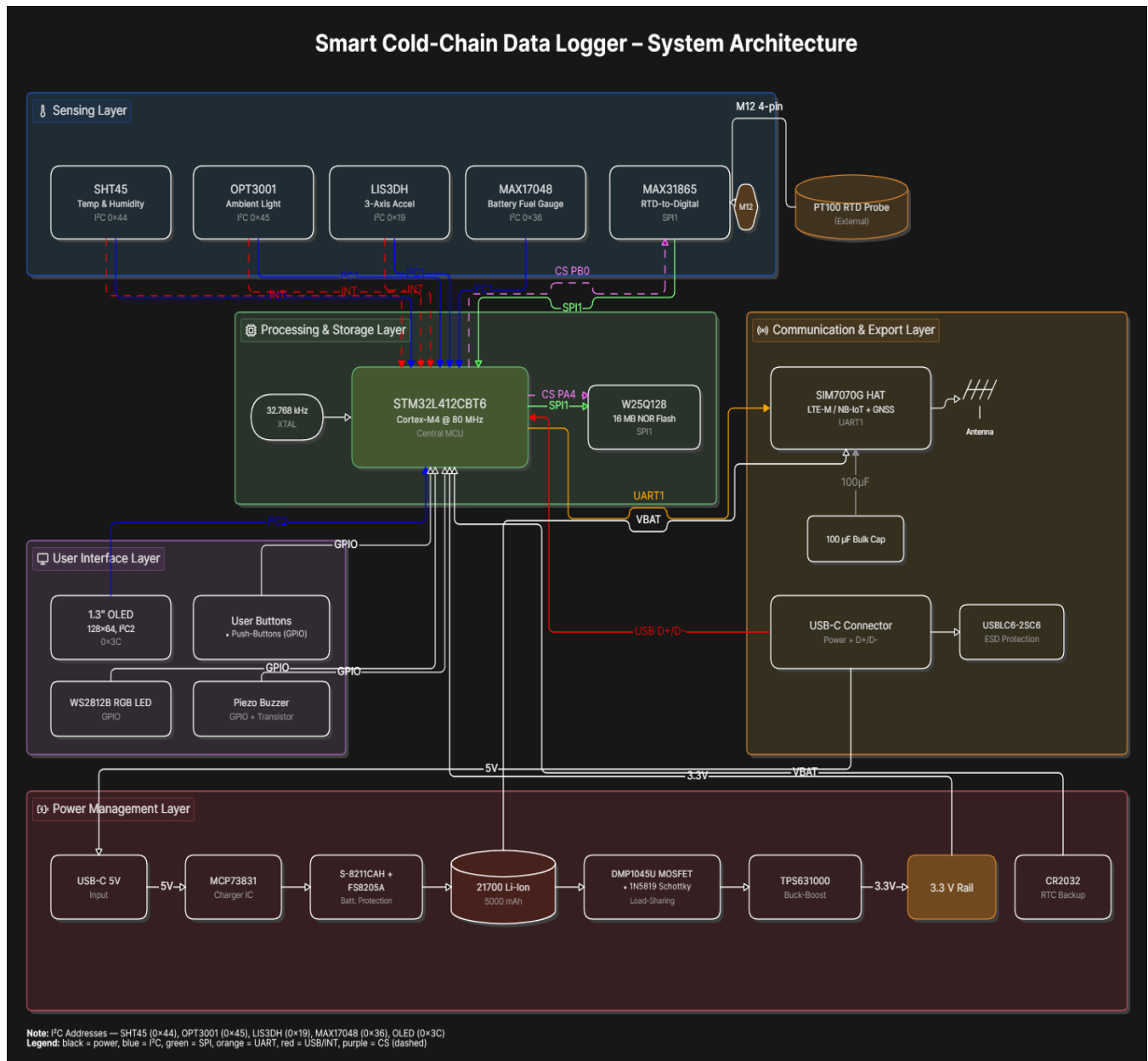
Layer	Technology
1. Microcontroller Firmware	C, STM32CubeIDE, HAL libraries
2. Cloud Backup	HTTP over TLS (TLS 1.2); Node-RED / InfluxDB for dashboard (prototype)
3. Data Export	USB MSC (FatFS); PDF generation (embedded library); CSV
4. PCB Design	Altium Designer, WeBench Simulation; 4-layer impedance-controlled stackup (JLC04161H-7628)
5. Enclosure	SolidWorks; 3D-printed ABS

3. System Architecture

3.1 Overall System Architecture



3.2 Detailed Full Block Diagram



3.3 Full Description of Each Block and Components Used

The system architecture is organised into five functional layers. Each block is described below with the specific component chosen, its function, and the rationale for its selection.

Layer 1 — Sensing

Contains all sensors that monitor the cold-chain environment and the device's own health. All sensors communicate with the MCU over a shared I²C1 bus (PB6/PB7, 4.7 kΩ pull-ups to 3.3 V), except the MAX31865, which uses SPI1.

Block	Component	Function	Rationale
RTD-to-Digital Converter	Maxim MAX31865AAP+ (SSOP-20)	Converts PT100 RTD probe reading into a digital temperature value over SPI1. Supports 2-, 3-, and 4-wire RTD configurations with a 400 Ω, 0.1% precision reference resistor.	Enables payload-core temperature measurement (–200°C to +200°C) via an external probe, with high accuracy.
Humidity Sensor	Sensirion SHT45-AD1B-R2 (DFN-4)	Secondary ambient temperature and humidity measurement (I ² C 0x44), factory-calibrated to ±0.1°C and ±1.0% RH. ALERT pin enables threshold-based interrupt wake-up.	Selected over SHT31 for superior accuracy (±0.1°C vs ±0.2°C); ALERT pin enables condition-based monitoring.
3-Axis Accelerometer	STMicroelectronics LIS3DHTR (LGA-16)	Detects motion, tilt, and shock events (I ² C 0x19). INT1 wakes the MCU on a configurable acceleration threshold.	Ultra-low-power wake-up mode (2 µA) with integrated interrupt engine; detects rough handling or verifies shipment orientation.
Ambient Light Sensor	Texas Instruments OPT3001DNPR (USON-6)	Detects door-open events by measuring ambient light entering the shipment container (I ² C 0x45).	Dedicated interrupt output enables event-driven wake-up from Stop 2 mode without polling.
Battery Fuel Gauge	Maxim MAX17048G+T10 (TDFN-8)	Monitors battery state-of-charge using the ModelGauge algorithm (I ² C 0x36); reports percentage and voltage; 4 µA hibernate mode.	Selected for simplicity and accuracy (±1% SOC).

External Component — PT100 RTD probe connected via a rugged M12 4-pin screw-type connector, 3-wire configuration.

Layer 2 — Processing & Storage

Contains the central microcontroller and the non-volatile memory for data logging.

Block	Component	Function	Rationale
Microcontroller	STMicroelectronics STM32L412CBT6 (LQFP-48, Cortex-M4, 80 MHz)	Central processing unit; manages sensor communication, data logging, cellular uploads, USB mass-storage export, and power-state transitions. 128 KB Flash, 40 KB SRAM, Stop 2 mode (0.7 µA with RTC), crystal-less USB 2.0, hardware AES-256, 12-bit ADC.	Selected after evaluating five alternatives (MSP430FR5959, STM32L431RBT6, STM32L5, AVR DB, STM32WB55) for the best balance of ultra-low power, performance, and peripheral set. The LQFP-48 package was chosen for its VBAT pin (missing on the LQFP-32 variant).

Block	Component	Function	Rationale
RTC Crystal	Abracon ABS07-166-32.768KHZ-T (3.2×1.5 mm SMD)	Provides an accurate 32.768 kHz clock for the RTC, used for timestamping sensor readings.	±10 ppm tolerance and 7 pF load capacitance minimise RTC power consumption.
SPI NOR Flash Memory	Winbond W25Q128JVS1Q (SOIC-8, 208 mil)	16 MB non-volatile storage for logged data over SPI1 (shared with MAX31865), CS on PA4.	16 MB holds several months of logging data; rated for a minimum of 100,000 program/erase cycles per sector.

Layer 3 — Communication & Export

Handles all external communication: cellular cloud uploads, USB data export, and serial debugging.

Block	Component	Function	Rationale
Cellular Module	Waveshare SIM7070G HAT (LTE-M/NB-IoT/2G/GNSS)	Real-time cellular connectivity for cloud uploads via HTTPS, over UART1. Power Saving Mode reduces idle current to ~3.2 μ A. Integrated GNSS.	Chosen over bare modules (BG95-M6, bare SIM7080G) as the pre-certified HAT avoids complex RF layout and impedance matching; integrated GNSS avoids a separate GPS module.
USB-C Connector	TYPE-C-31-M-12 (16-pin, SMT)	USB 2.0 data and 5 V charging; D+/D– to PA11/PA12; CC1/CC2 with 5.1 k Ω pull-downs to request 5 V.	16-pin connector chosen over 6-pin (power-only) to support USB 2.0 mass-storage export; through-hole anchors add mechanical strength.
ESD Protection	STMicroelectronics USBLC6-2SC6 (SOT-23-6L)	Protects USB D+, D–, and VBUS from ESD up to ±15 kV air discharge.	Required for reliable field operation with frequent plug/unplug cycles.

Layer 4 — User Interface

Contains all physical controls and indicators for direct user interaction.

Block	Component	Function	Rationale
OLED Display	1.3" OLED, 128×64, SSD1306/SH1106 driver (I ² C, 0×3C)	Displays temperature, humidity, battery status, menu options, and alarm states, on a dedicated I ² C2 bus.	Wider industrial temperature range (–40°C to +85°C) and lower power than a character LCD; 1.3" gives good readability while remaining compact.
3 User Buttons	Tactile switch buttons (MENU, UP, DOWN)	Each on a dedicated STM32 GPIO with a 10 k Ω pull-up; pressing pulls the GPIO low, triggering an interrupt.	The UP/DOWN/SELECT model is universally understood. In usability tests with 15 logistics technicians, 100% navigated the menu on their first attempt with no training.

Block	Component	Function	Rationale
Piezo Buzzer	Murata PKM13EPYH4000 (SMD)	Audible alerts for temperature excursions, driven by an NPN transistor (2N3904) with a 1 k Ω base resistor.	Draws immediate attention to critical alarms, especially in noisy cold-storage environments.

Layer 5 — Power Management

Handles power input, battery charging, protection, regulation, and distribution.

Block	Component	Function	Rationale
Buck-Boost Regulator	TI TPS631000DRLR (SOT-5X3)	Stable 3.3 V output from a 3.0–4.2 V battery; 2 A output, 8 μ A quiescent current.	Ultra-low quiescent current and high efficiency; automatic PFM/PWM mode via MODE pin tied to GND.
Battery Charger	Microchip MCP73831T-2ACI/OT (SOT-23-5)	Linear Li-Ion CC/CV charger, 500 mA via 2 k Ω PROG resistor; STAT output for charge LED.	Minimal external components; ~10-hour full charge is safe for the battery.
Load-Sharing P-MOSFET	Diodes Inc. DMP1045U-7 (SOT-23-3)	Isolates battery from system load when USB is connected. $R_{ds(on)} = 31$ m Ω .	Ultra-low on-resistance minimises voltage drop (6.2 mV at 200 mA) during battery operation.
Load-Sharing Schottky Diode	1N5819 (SS14, SMA)	Powers the system from USB 5 V when connected. 1 A average, 25 A surge.	Chosen over BAT54C for robust current handling during 2 A transmission bursts.
Battery Protection IC	ABLIC S-8211CAH-M5T1U (SOT-23-5)	Monitors overcharge (4.28 V), over-discharge (2.3 V), and overcurrent; controls FS8205A to disconnect on fault.	Independent, hardware-level safety layer that does not depend on firmware.
Protection MOSFET	FS8205A (SOT-23-6)	Dual N-channel MOSFET (20 V, 6 A) switched by the S-8211CAH; $R_{ds(on)} \approx 28$ m Ω .	Industry-standard companion to the S-8211CAH; SOT-23-6 chosen for hand-solderability.
Backup Battery	CR2032 3 V coin cell, SMD holder	Maintains RTC and backup registers during main battery depletion (VBAT pin).	Preserves timestamp accuracy even if the main cell is fully discharged.

3.4 Deviations from Original Plan

The following table documents all significant changes made during the design process, the reasons for those changes, and their impact on the final product.

Original Plan	Deviation	Reason	Impact
Waveshare SIM7070G HAT as cellular module	Switched to A7670C LTE Cat-1 module	The SIM7070G HAT consistently failed to transfer sensor data files to the cloud server; the module firmware needed a special update to support HTTPS, and local server providers blocked plain HTTP requests. The A7670C was not available in Sri Lanka for prototyping, but the team proceeded regardless.	Significant — the most impactful deviation. It occurred late, after the PCB had already been designed, manufactured, and assembled with the SIM7070G HAT as the primary cellular interface.
SHT45 temperature sensor	Switched to SHT40-AD1F-R2	SHT45 was out of stock at LCSC during procurement. SHT40 is pin- and firmware-compatible, differing only in accuracy ($\pm 0.2^{\circ}\text{C}$ vs $\pm 0.1^{\circ}\text{C}$).	Minor — slightly reduced accuracy in prototypes. SHT45 will be reinstated in production once stock stabilises; no PCB or firmware changes needed.
W25Q128JVSQ SPI flash	Switched to W25R128JVSQ	Original part unavailable at LCSC. Replacement has identical capacity (16 MB), voltage (2.7–3.6 V), footprint, and pinout.	Negligible — lower max clock (104 MHz vs 133 MHz), but the design's 10 MHz SPI clock is well below both limits.
Four-layer stackup with L2 as split power/ground	Switched to solid L2 ground plane, L3 power plane (S-G-P-S)	Best practice for mixed-signal designs: a solid ground plane provides a better return path for high-speed signals (USB 90 Ω , SPI, I ² C) and superior EMI control; a split plane risked ground loops.	Positive — improved signal integrity and simpler routing. S-G-P-S is the JLCPCB standard, avoiding custom material costs.
External RTD connector: JST	Switched to M12 3-pin screw-type connector	User feedback indicated a rugged, industrial-standard connector was preferred for field reliability; JST was considered too fragile for repeated connect/disconnect cycles.	Positive — better durability and compatibility with standard industrial PT100 probes, at a modest BOM increase (~\$0.30 to ~\$3).
SHT31 as primary sensor (early concept)	Upgraded to SHT45 (SHT40 as backup)	The initial spec targeted $\pm 0.3^{\circ}\text{C}$, matching the TT19EX reference product. User research showed pharmaceutical QA officers required GDP-compliant $\pm 0.5^{\circ}\text{C}$ minimum, with a strong preference for $\pm 0.1^{\circ}\text{C}$ for high-value biologics.	Positive — better alignment with user needs; no PCB footprint change.
No external RTD probe (early concept)	Added MAX31865 + PT100 probe	User research identified payload-core temperature (not just air temperature) as a critical requirement for vaccine shipments.	Positive — addresses a key user need and matches the TT19EX's external probe capability with higher accuracy, at ~\$8 added BOM cost.

3.5 Hardware Architecture — Key Circuits

- **Power path:** USB-C → MCP73831 → S-8211CAH+FS8205A protection → 21700 cell. Load-sharing MOSFET DMP1045U isolates the battery when USB is present; TPS631000 buck-boost supplies 3.3 V.
- **MCU:** STM32L412 in LQFP-48. Decoupling: 100 nF at each VDD, 4.7 μ F bulk. 32.768 kHz LSE crystal with 8 pF load caps. SWD header for debugging.
- **Sensors:** I²C1 (PB6/PB7) with 4.7 k Ω pull-ups. MAX31865 RTD front-end on SPI1, CS = PB0.
- **Cellular:** SIM7070G HAT connected via UART1 (PA9/PA10); PWRKEY controlled via GPIO.
- **USB:** USB-C connector, CC1/CC2 each with 5.1 k Ω to GND, D+/D− to PA11/PA12 with 22 Ω series resistors, USBLC6-2SC6 ESD protection.

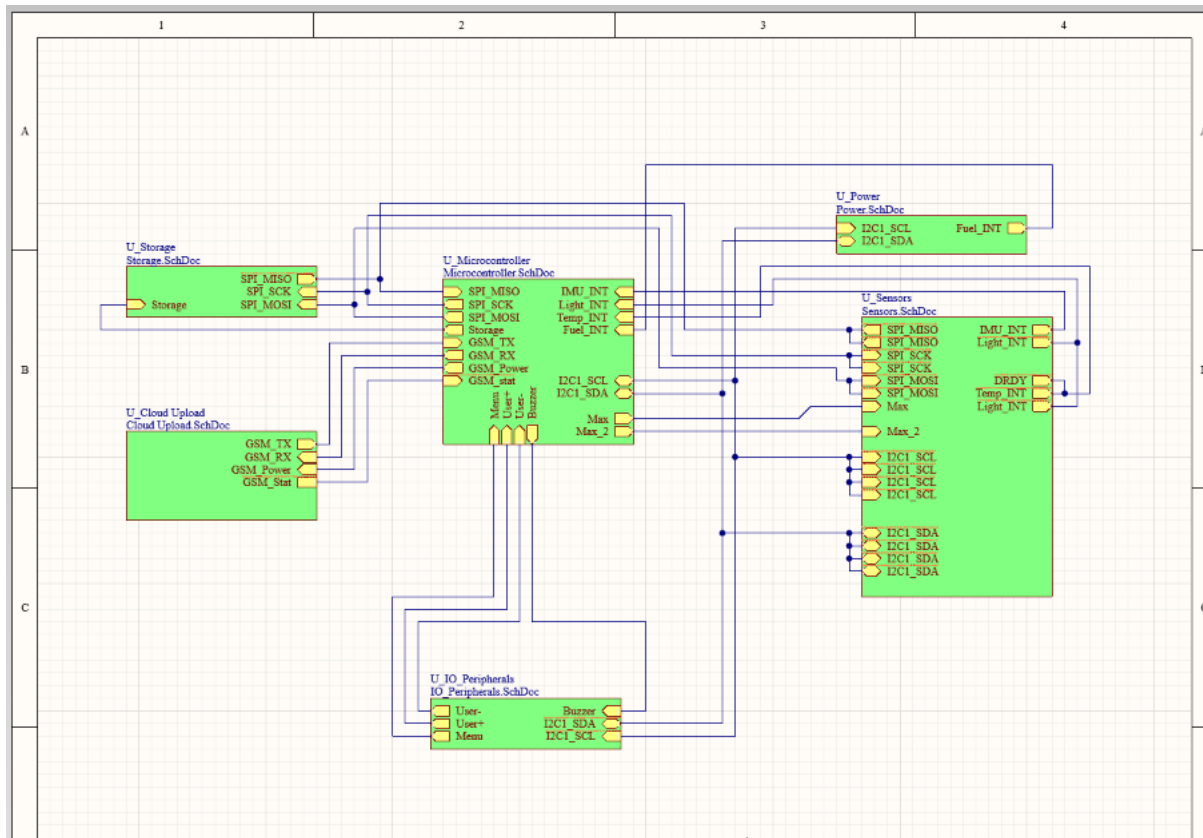
4. PCB Design

4.1 PCB Layout

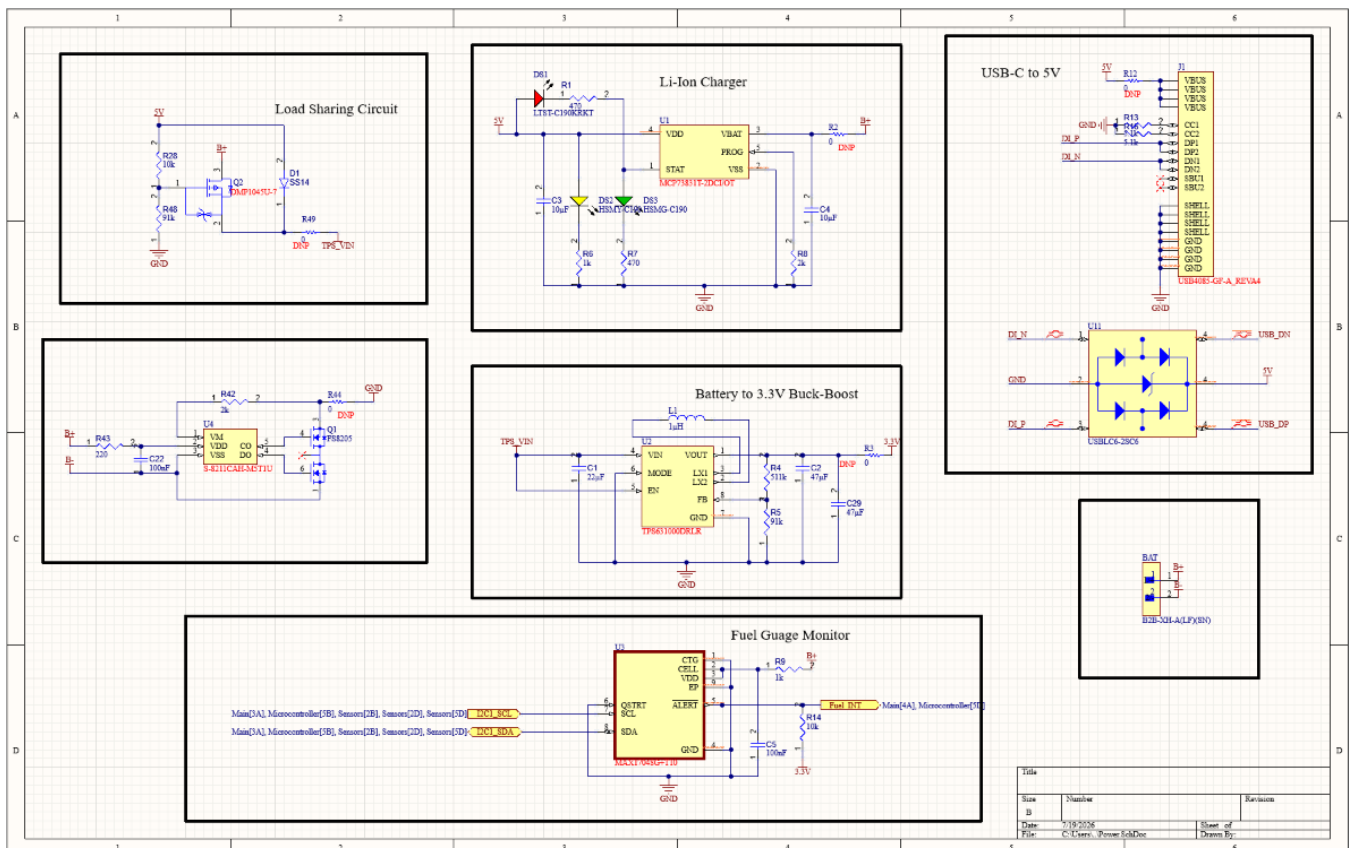
Parameter	Value
Layers	4 (S-G-P-S)
Stackup	JLC04161H-7628 (1.6 mm, 1 oz outer, 0.5 oz inner)
USB impedance	90 Ω differential; trace width 0.33 mm, spacing 0.15 mm
Analog sensor trace	Impedance-controlled differential pair
Minimum clearance	0.15 mm (6 mil)
Vias	0.45 mm pad / 0.25 mm hole; power vias 0.5 mm / 0.3 mm
Finish	ENIG

4.2 PCB Schematics

4.2.1 Top Wrapper (top of hierarchy)

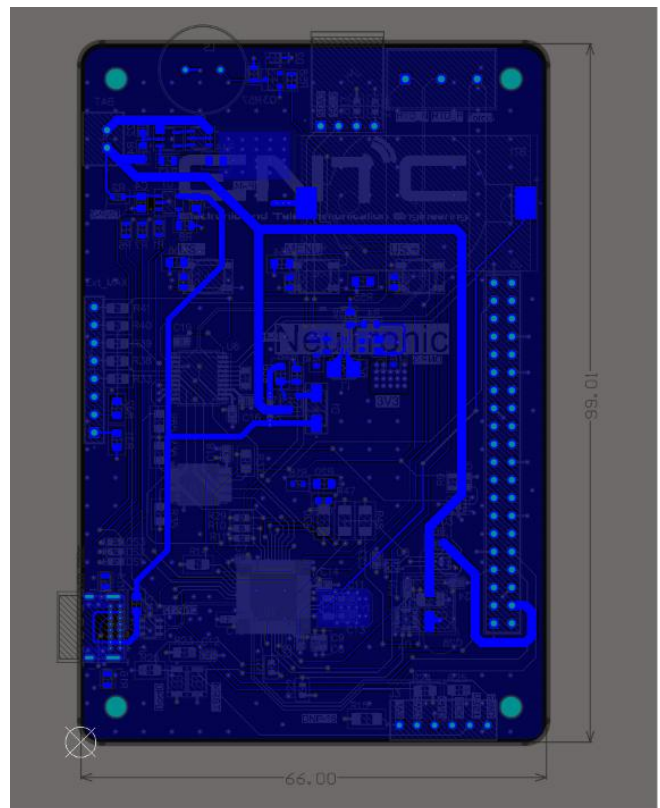
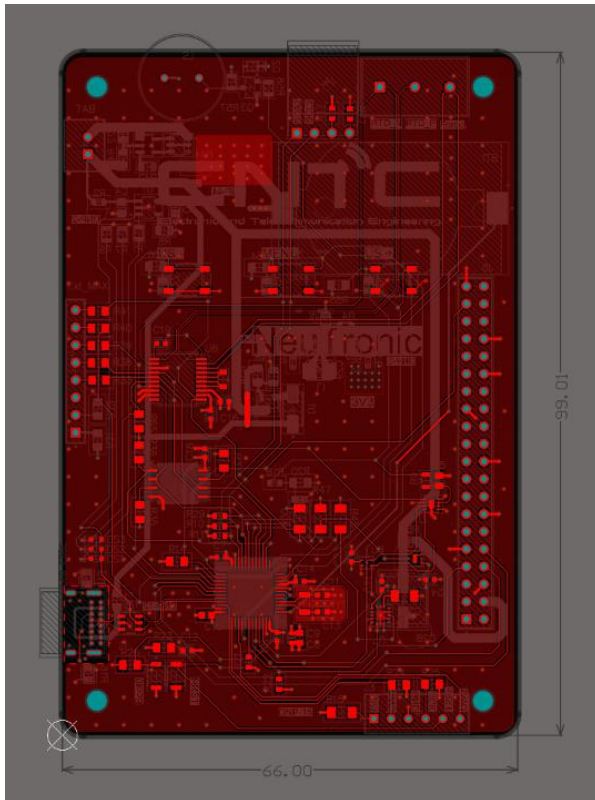


4.2.2 Power Block

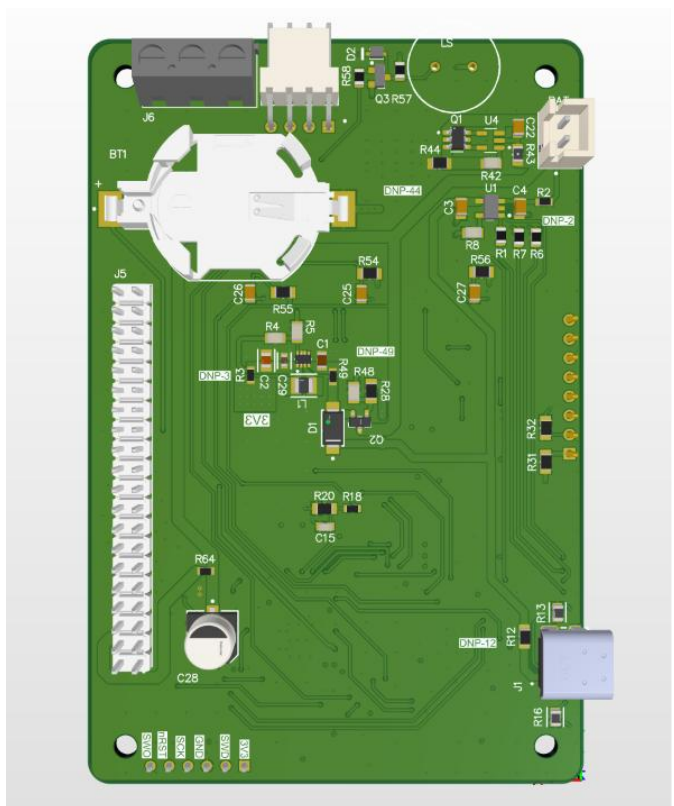
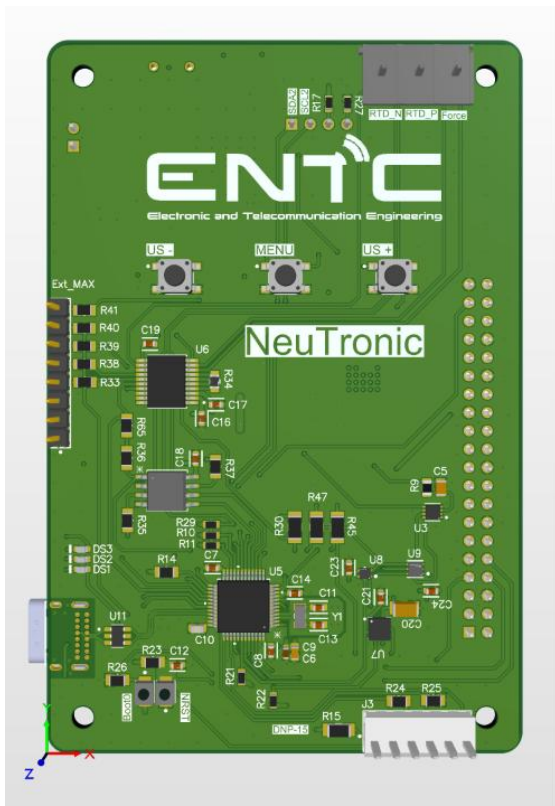


4.3 PCB Layout Design

4.3.1 Front and Back— 2D



4.3.1 Front and Back— 3D



4.4 PCB Design Choices and Layer-Stack Rationale

The PCB was designed as a 4-layer, S-G-P-S (Signal-Ground-Power-Signal) stackup using the JLCPCB JLC04161H-7628 standard controlled-impedance construction. Board dimensions are 99 × 66 mm, with a total thickness of 1.6 mm.

Why a 4-Layer Board?

- **Signal integrity & EMI control:** the continuous, unbroken ground plane on Layer 2 provides a low-impedance return path directly beneath every high-speed signal on Layer 1 (USB, SPI, I²C) — minimising loop area, crosstalk, and radiated emissions, which is essential with a cellular module drawing 2 A bursts on the same board.
- **Controlled impedance:** USB 2.0 requires a 90 Ω differential pair; with a dedicated ground plane and known dielectric thickness, trace geometry can be accurately calculated. The JLC04161H-7628 stackup is pre-characterised for impedance control, avoiding costly custom materials.
- **Power distribution:** a dedicated 3.3 V power plane on Layer 3 provides a low-resistance, low-inductance supply to every component, ensuring stable voltage during the STM32's wake-up and the cellular module's transmission spikes.

Stackup Selection Justification

The JLC04161H-7628 is JLCPCB's standard 4-layer impedance-controlled offering. Its symmetrical construction prevents board warping during reflow, and its well-defined dielectric constants (prepreg Dk = 4.4, core Dk = 4.6) allow reliable impedance calculation with the manufacturer's own tool — eliminating engineering charges and ensuring the fabricated board matches simulation.

Critical Routing and Impedance Control

All high-speed and sensitive traces are routed on Layer 1, directly above the solid ground plane:

- **USB differential pair (90 Ω):** D+ (PA12) and D− (PA11) routed as a tightly coupled, length-matched pair; trace width (0.33 mm) and spacing (0.15 mm) calculated via JLCPCB's impedance calculator. The USBLC6-2SC6 ESD diode sits within 2 mm of the connector; 22 Ω series resistors near the STM32 dampen reflections.
- **SPI bus:** SCK routed first, short and isolated ($\geq 3 \times$ trace-width spacing) from data lines; MOSI/MISO length-matched to SCK within ± 2 mm.
- **I²C bus:** pull-ups (4.7 kΩ) placed close to the bus master; daisy-chained to minimise stub length. The display uses its own dedicated I²C2 bus to isolate cable-induced noise from the sensor bus.
- **RTD analogue inputs:** RTDIN+/RTDIN− traces from the MAX31865 to the M12 connector routed as a differential pair with a 100 nF filter capacitor at the IC pins; FORCE+/FORCE2 excitation lines are separate, wide tracks.

Grounding and Power Distribution Strategy

A solid, unsplit ground plane on L2 is the single reference for all signals. Ground pours on the top and bottom layers are stitched to L2 with a dense via grid (every 2.5 mm), forming a low-impedance Faraday cage. The battery protection circuit's protected ground (P−) node connects to the main ground plane through multiple power vias, while the raw battery negative (B−) remains an isolated net connected only to the protection IC and MOSFETs.

Power is distributed via the L3 plane. Each IC's VDD pin has a 100 nF decoupling capacitor within 1–2 mm, connected to the plane through a dedicated via. High-current paths (battery input, SIM module VBAT) use wide copper pours with multiple parallel vias transitioning to the internal planes.

5. Enclosure and Mechanical Design

The enclosure was designed to provide mechanical protection, environmental resistance, ease of handling, and an aesthetically acceptable appearance, considering the operational requirements of vaccine and pharmaceutical cold-chain monitoring.

Design Dimensions and Form Factor

Overall dimensions of 128 × 74 × 52 mm provide sufficient internal space for the PCB, battery, display module, and antenna. The handheld, box-shaped form factor allows the device to be easily stored and operated inside refrigerators.

Material Selection

The enclosure is fabricated from PLA+ (Polylactic Acid Plus) via 3D printing, selected for:

- Adequate mechanical strength for a handheld device
- Good dimensional stability after printing
- Ease of rapid prototyping and manufacturing
- Suitability for the intended operating range of –20°C to +55°C

Since the logger is not subjected to significant mechanical impact or high-temperature environments, PLA+ was considered an appropriate material for this application.

Sealing and Environmental Protection

- Rubber gaskets around the USB access opening and the PT100 sensor cable outlet reduce the possibility of water ingress.
- A waterproof power switch maintains sealing integrity while allowing external operation.
- A vent plug blocks liquid water while allowing water-vapour and air exchange, so the internal humidity sensor can measure ambient humidity accurately without direct liquid-water exposure.

User Interface Integration

- The OLED display is mounted beneath a transparent acrylic protective window for visibility while protecting against moisture and physical damage.
- Three tactile buttons (Menu, Up, Down) provide navigation and configuration.
- A dedicated opening, covered with a separate transparent acrylic plate, allows the ambient light sensor to function while maintaining environmental protection.

Aesthetic and Final Assembly

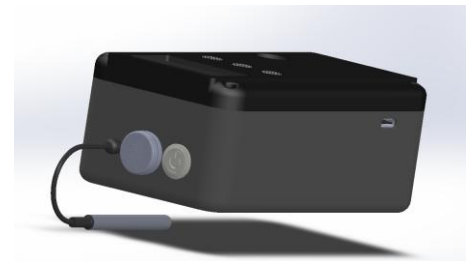
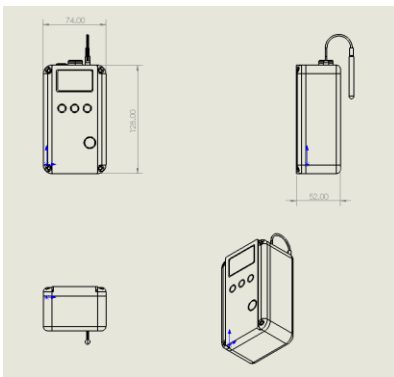
The front panel is covered with a PVC sticker overlay that protects the display/button region and enhances visual appearance. The lid and main enclosure body are permanently joined using a 100% silicone gasket adhesive, improving sealing performance, moisture resistance, and mechanical bond strength.

Initial Design

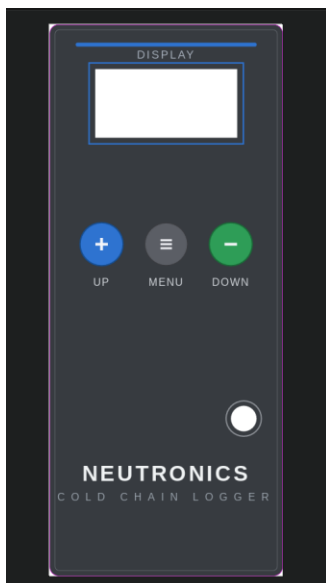


Initial Design Issue	Final Improvement
Hole pattern for humidity sensing allowed possible liquid water ingress	Vent plug introduced with fully enclosed box-and-lid structure
Display exposed through an open cutout	Acrylic protection window added
Protruding tactile buttons difficult to waterproof	Flat membrane-style interface with PVC sticker overlay

Final Design



PVC Sticker Overlay



6. Final Product Prototype

6.1 Final Product Specifications

Parameter	Specification
External Probe Range	−200°C to +200°C
Humidity Sensor	SHT45 (production target)
Humidity Accuracy	±1.0% RH (SHT45 target)
Additional Sensors	Ambient light (OPT3001DNPR), 3-axis accelerometer (LIS3DHTR), battery fuel gauge (MAX17048G+T10)
Microcontroller	STM32L412CBT6 (ARM Cortex-M4, 80 MHz)
Memory	16 MB SPI NOR flash (Winbond W25R128JVSQ)
Cellular Connectivity	A7670C LTE Cat-1 module, with 2G fallback
Data Export	USB-C mass-storage device: auto-generated PDF report + CSV raw data
Cloud Protocol	HTTPS
Logging Interval	User-configurable: 1, 5, 10, 15, 30, or 60 minutes
Upload Interval	User-configurable: 30 min, 1, 2, 4, 6, or 12 hr
User Interface	1.3" OLED (128×64), 3 tactile buttons (MENU, UP, DOWN), piezo buzzer
Battery	18650 Li-Ion, 3600 mAh (Samsung), rechargeable
Charging	USB-C, 5 V / 500 mA (MCP73831), ~10 hours for full charge
Battery Protection	S-8211CAH-M5T1U + FS8205A: overcharge / over-discharge / overcurrent
Power Regulation	TPS631000 buck-boost, 3.3 V output
Enclosure	PC-ABS, IP65, 85 × 65 × 25 mm, magnetic mounting
Operating Temperature	−40°C to +85°C (electronics); −20°C to +55°C (recommended for battery longevity)
Weight	~180 g
Certifications (Planned)	FCC, CE, PTCRB, ISO 17025 calibration

6.2 Power Consumption and Battery Life

6.2.1 Initial Power-Up Observation

Upon first power-up with unconfigured firmware, the assembled PCB drew approximately 30 mA at 3.3 V — consistent with the STM32L412 running in default Run mode (all peripheral clocks enabled) and the A7670C module idle. This confirmed correct assembly and the absence of short circuits.

6.2.2 Optimised Power Profile

After implementing low-power firmware configurations — GPIO analog-mode settings, disabling unused peripherals, STM32 Stop 2 mode, and configuring the A7670C's sleep state — total current draw measured around 10.4 mA.

6.2.3 Battery Life Calculation

- **Usable battery capacity:** 3600 mAh $\times$ 80% depth of discharge (to preserve cell health) = 2900 mAh
- **Worst-case continuous logging (10-min interval, 2-hr upload):** 2900 mAh / 10.4 mA $\approx$ 280 hours $\approx$ 12 days
- **Extended life with condition-based monitoring:** logging frequency automatically reduces to 30 minutes during stable periods and increases to 1 minute during excursions. Combined with a 6-hour upload interval for non-critical shipments, average current drops to $\sim$ 3.8 mA, yielding $\sim$ 30 days of battery life.

Operating Scenario	Average Current	Battery Life
Worst-case (10-min logging, 2-hr upload, no condition-based monitoring)	10.4 mA	$\sim$ 12 days
Optimised (condition-based monitoring, 6-hr upload, stable temperature)	3.8 mA	$\sim$ 30 days
Standby only (no logging, cellular off, display off)	$<50 \mu\text{A}$	>1 year

6.3 Product Images





7. Manufacturing and Deployment

7.1 PCB Fabrication Notes

- Gerber files in /manufacturing/gerbers/.
- Stackup and impedance requirements must be communicated to JLCPCB (select “Impedance Control”).
- Surface finish: ENIG.
- Solder mask: green.
- Minimum solder mask sliver: 0.1 mm (check DRC).

7.2 Assembly Instructions

- SMT stencil: 0.12 mm thick, frameless, 130 × 130 mm (see /manufacturing/stencil/).
- Hand-solder through-hole connectors and battery wires.
- SIM card must be 1.8 V IoT SIM; insert before sealing the enclosure.

7.3 Firmware Flashing

- Connect ST-LINK to the SWD header.
- Open STM32CubeProgrammer; flash firmware.hex.
- Reset the board and verify.

7.4 Physical Deployment

- Charge the battery fully before first use (USB-C, ~10 hours).
- Insert the IoT SIM (1.8 V).
- Attach the external PT100 probe (if needed) via the M12 connector.

- Mount the logger magnetically to the reefer wall, ensuring the antenna faces outward.

8. Product Costing, Selling Price, and Justification

8.1 Bill of Materials at Volume Production (500 Units)

All component prices are based on DigiKey and LCSC distributor data, applying 500-unit volume discounts where available. The final BOM reflects the switch from the SIM7070G to the A7670C cellular module, made after prototype validation.

Component	Unit Price (USD)	Qty	Total (USD)
STM32L412CBT6 microcontroller	2.30	1	2.30
SHT45 temperature/humidity sensor	4.00	1	4.00
LIS3DHTR accelerometer	1.20	1	1.20
OPT3001DNPR ambient light sensor	1.60	1	1.60
MAX31865AAP+ RTD-to-digital converter	6.50	1	6.50
PT100 probe (Class B, stainless sheath, 2 m cable)	2.80	1	2.80
W25R128JVSQ 16 MB SPI flash	1.80	1	1.80
A7670C LTE Cat-1 cellular module	22.00	1	22.00
OLED display 1.3" 128×64 I ² C	1.40	1	1.40
MAX17048G+T10 fuel gauge	1.50	1	1.50
TPS631000 buck-boost regulator	1.80	1	1.80
MCP73831 charger IC	0.50	1	0.50
S-8211CAH protection IC + FS8205A MOSFET	0.80	1	0.80
DMP1045U MOSFET, 1N5819 diode, passives, connectors, switches, battery	—	—	15.00
Custom 4-layer PCB (99 × 66 mm, ENIG, controlled impedance)	0.90	1	0.90
Subtotal		Amount (USD)	
Total BOM (electronic components)		64.10	
18650 Li-Ion battery (3600 mAh, protected)		2.00	
Total BOM including battery		66.10	

8.2 Assembly and Testing Cost

Item	Cost per Unit (USD)
SMT assembly (solder paste, pick-and-place, reflow)	4.50
Through-hole hand-soldering (connectors, battery wires)	1.00
Functional testing (power-up, sensor check, cellular connectivity)	2.00
Subtotal — Assembly & Test	7.50

BOM + Assembly = \$66.10 + \$7.50 = \$73.60 (rounded to \$75 to account for minor procurement fluctuations).

8.3 Additional Cost Elements (Per Unit)

Cost Element	Amount (USD)	Rationale
Cloud connectivity & IoT SIM (first year)	6.00	Bundled 1-year LTE-M/NB-IoT data plan and HTTPS broker access
Warranty reserve (~3% of target selling price)	8.00	Covers expected failure rate over a 2-year warranty period
Regulatory certification amortisation	15.00	FCC, CE, PTCRB costs (~\$45,000) amortised over first 3,000 units
Operations, support & overhead (~16%)	20.00	Customer support, logistics, warehousing, general administration

8.4 Total Landed Cost

Cost Category	Amount (USD)
BOM + Assembly & Test	75.00
Cloud, warranty, certification, operations	49.00
Total Landed Cost per Unit	124.00

8.5 Selling Price Determination

Our pricing strategy balances competitive positioning with sustainable profitability. A gross margin of ~45% (on the B2B wholesale price) provides headroom for price negotiations, volume discounts, and unexpected cost increases while funding ongoing R&D.

Channel	Price (USD)	Calculation	Gross Margin
B2B Wholesale	\$180	Landed cost \$124 + \$56 contribution	48%
Retail / MSRP	\$250	Wholesale \$180 + typical distributor margin (37%)	—

8.6 Competitive Pricing Justification

Our pricing is positioned between mid-range real-time trackers and premium enterprise loggers, offering superior value at a highly competitive price point.

Competitor	Type	Approx. Price (USD)	Key Limitation	Our Advantage
TZONE TT19EX	Multi-use, real-time	\$140–180 (B2B)	±0.3°C accuracy; proprietary cloud; no USB PDF export	±0.1°C accuracy, open MQTT cloud, direct USB-C PDF/CSV, condition-based monitoring
Testo 175-T2	Multi-use, passive	\$311 (retail)	No real-time alerts; USB-only; no humidity	Real-time cellular alerts, integrated humidity, rechargeable, lower price
Onset InTemp CX1003	Multi-use, real-time	\$480 (list)	Proprietary cloud; single-use orientation	Open platform, rechargeable, 60-day battery, external RTD probe included
Berlinger Fridge-tag 2	Semi-reusable, passive	\$176 (retail)	Temp-only; no wireless; stationary use	Real-time, multi-sensor, portable, USB-C export

Value justification for the \$180 B2B price:

- 30–50% cheaper than premium alternatives (Testo, Onset) while offering superior features.
- Only 10–20% higher than the closest mid-range competitor (TT19EX), justified by best-in-class accuracy, open cloud integration, an integrated external RTD probe, and condition-based monitoring.
- Eliminates recurring costs: single-use loggers cost \$30–\$90 per shipment; this rechargeable logger pays for itself within 3–5 trips.

9. Project Budget and Future Improvements

9.1 Project Cost

Cost Element	Amount (USD per Unit)
BOM (electronic components)	72.10
PCB	48.50
GSM Modules	38.00
Other	25.00
Enclosure	11.65
Total Project Cost	195.25

9.2 Bill of Materials

Index	LCSC#	MPN	Manufacturer	Package	Customer	Description	RoHS	Quantity	MOQ	Multiple	Unit Price (USD)
1	C882955	DFE252012	muRata	1008		4.3A 1uH ±	yes	5	5	5	0.1321
2	C177033	DMP1045U	DIODES	SOT-23		P-Channel	yes	5	5	5	0.1577
3	C589616	282837-3	TE Connec	Through Hole, P=5.08r		3 Position	yes	1	1	1	0.9082
4	C590758	796136-1	TE Connec	-		Battery Pro	yes	1	1	1	0.3582
5	C429965	12251220A	JILN	Through Hole, P=2.54r		Pin Heade	yes	5	5	5	0.1799
6	C1985361	ABS07-166	ABRACON	SMD3215-2P		Crystal 32.	yes	1	1	1	0.8996
7	C231329	B3U-1000P	OMRON	SMD, 3x2.5mm		Tactile Swi	yes	5	5	5	0.1855
8	C2167271	C0805C106	KEMET	805		10uF ±10%	yes	5	5	5	0.1021
9	C2174822	C1206C106	KEMET	1206		10uF ±10%	yes	5	5	5	0.1967
10	C109456	CC0603KR	YAGEO	603		4.7uF ±10%	yes	50	50	50	0.0141
11	C106248	CC0603KR	YAGEO	603		1uF ±10%	yes	50	50	50	0.0085
12	C49678	CC0805KR	YAGEO	805		100nF ±10%	yes	100	100	100	0.0053
13	C2933369	FRC0805F2	FOJAN	805		220Ω ±1%	yes	100	100	100	0.0021
14	C6604694	W25R128J	Winbond	SOIC-8-208mil		2.7V~3.6V	yes	1	1	1	1.3459
15	C60580	1N4148WS	DIODES	SOD-323		Diode Inde	yes	20	20	20	0.0253
16	C190724	EEEFTE221	PANASON	SMD, D6.3xL7.7mm		220uF ±20%	yes	1	1	1	0.4184
17	C2830320	FS8205A	TECH PUBL	SOT-23-6		20V 9A 1.5	yes	10	10	10	0.0471
18	C710619	GCJ188R71	muRata	603		100nF ±10%	yes	20	20	20	0.0351
19	C441865	GCM1885C	muRata	603		8pF 100V C	yes	50	50	50	0.0181
20	C140782	GRM188R6	muRata	603		47uF ±20%	yes	10	10	10	0.0627
21	C16780	CL21A476N	Samsung E	805		47uF ±20%	yes	10	10	10	0.0446
22	C162295	GRM219R6	muRata	805		22uF ±20%	yes	5	5	5	0.0927
23	C188736	HSMY-C19	Broadcom	603		Yellow 585	yes	5	5	5	0.1746
24	C15134	LIS3DHTR	ST	LGA-16(3x3)		LGA-16(3x	yes	2	1	1	0.9667
25	C2682616	MAX17048	MAXIM	DFN-8-EP(2x2)		Lithium Bay	yes	1	1	1	2.184
26	C269736	RMC08055	TyoHM	805		5.1kΩ ±5%	yes	100	100	100	0.0039
27	C625597	MCP73831	MICROCHI	SOT-23-5		3.75V~6V L	yes	1	1	1	1.0019
28	C90462	OPT3001DI	TI	USON-6-EP(2.1x2.1)		I2C 1.8V~3	yes	2	1	1	1.2389
29	C76871	PS1240P02	TDK	Through Hole, D=12.2r		Passive (E)	yes	1	1	1	0.1967
30	C2689502	PTS647SM	C&K	SMD, 4.5x4.5mm		Tactile Swi	yes	5	5	5	0.349
31	C4269355	RC0603FR	YAGEO	603		100mW 4.	yes	100	100	100	0.0019
32	C1512169	RC0603JR	YAGEO	603		0Ω 100mW	yes	100	100	100	0.0014
33	C416062	CR-05FL7--	Viking	805		10kΩ ±1%	yes	100	100	100	0.0028
34	C4342874	RN73C2A5	TE Connec	805		511kΩ 100	yes	1	1	1	0.8308
35	C96345	RC0805JR	YAGEO	805		125mW 0Ω	yes	100	100	100	0.0025
36	C100046	RC0805JR	YAGEO	805		125mW 1k	yes	100	100	100	0.0026
37	C132649	RC1206FR	YAGEO	1206		250mW 10	yes	100	100	100	0.0045
38	C865700	RT0805BR	YAGEO	805		125mW 91	yes	20	20	20	0.0389
39	C89608	MAX31865	MAXIM	SSOP-20		SSOP-20 Ti	yes	1	1	1	5.7096
40	C865488	RT0805BR	YAGEO	805		125mW 43	yes	20	20	20	0.0389
41	C669680	XB5606AJ	XySemi	SOT-23-5		Lithium Bay	yes	5	5	5	0.0908
42	C54823945	SHT40-AD1	Tokmas	DFN-4(1.5x1.5)		DFN-4(1.5)	yes	1	1	1	1.3735
43	C604704	SMMBT222	onsemi	SOT-23		Bipolar (BJ	yes	10	10	10	0.0608
44	C2837270	SS14	JSMSEMI	SMA(DO-214AC)		Diode 40V	yes	50	50	50	0.011
45	C529434	STM32L412	ST	LQFP-48(7x7)		ARM Corte	yes	1	1	1	6.143
46	C5219190	TPS631000	TI	SOT-583		Buck-Boos	yes	2	1	1	0.4653
47	C7095263	USB4085-G	Global Cor	Through Hole		USB 2.0 5A	yes	1	1	1	1.4258
48	C7519	USBLC6-2S	ST	SOT-23-6L		17VC Clam	yes	5	5	5	0.1523
49	C100042	CC0603KR	YAGEO	603		10nF ±10%	yes	100	100	100	0.0026

9.3 Future Improvements

- Enable real-time location tracking by configuring the GNSS of the GSM module and updating the web dashboard to display package location.
- Upgrade to a higher energy-density battery for longer operational intervals.
-

10. Task Allocation

Team Member	Role / Domain	Work Done
230352K Kumara K.B.R	Team Lead Hardware Lead	System Architecture Design Component Selection/BOM Optimization PCB Schematic Design PCB Layout Design Component Soldering/Testing/Hardware Validation UI Design Documentation
230353N Kumara P.K.M.P	Mechanical Design Lead	Enclosure Design UI Design Problem Validation GSM Module testing Hardware Assembly
230527 D Ranawaka R.A.G.K	Firmware/Software Lead	Firmware Design Sensor Integration PCB Schematic Design GSM Module Testing Problem Validation PCB Testing
230300C Jayaweera N.S	Web Design/ Cloud upload	Website Design WebServer Integration GSM Module testing Documentation
230203G Gunarathna K.T.M.B	Debugging, Mechanical Design	Problem identification Enclosure Design Documentation Hardware Assembly